

ident, **x**, **y**, **this**
location, ℓ
class_name, **C**, **D**, **E**
field_name, **f**
method_name, **m**
constr_name, **K**
index, h , i , j , k , n

<i>class_def</i>	$::=$ $ \quad \mathbf{class\,C\,extends\,D}\{C_1\,f_1, \dots, C_i\,f_i; K, M_1, \dots, M_j\}$	
<i>constr_def, KK</i>	$::=$ $ \quad C(C_1\,x_1, \dots, C_i\,x_i, C'_1\,x'_1, \dots, C'_j\,x'_j)\{\mathbf{super}\,(x_1, \dots, x_k); f_1 = x'_1, \dots, f_h = x'_h; \}$	
<i>method_def, M</i>	$::=$ $ \quad C\mathbf{m}(C_1\,x_1, \dots, C_i\,x_i)\{\mathbf{return\,e}; \}$	
<i>expr, e</i>	$::=$ $ \quad \mathbf{x}$ $ \quad \mathbf{e.m(e_1, \dots, e_n)}$ $ \quad \mathbf{e.f}$ $ \quad \mathbf{new\,C(e_1, \dots, e_n)}$ $ \quad (C)\mathbf{e}$ $ \quad (e)$ $ \quad \{e_1/x_1, \dots, e_k/x_k, \mathbf{this/e'}\}e''$	M M
<i>terminals</i>	$::=$ $ \quad \longrightarrow$ $ \quad \rightarrow$ $ \quad \mathbf{extends}$ $ \quad \mathbf{return}$ $ \quad =$ $ \quad <:$ $ \quad \nlessdot:$ $ \quad \neq$ $ \quad \vdash$	
Γ	$::=$ $ \quad \mathbf{x : C}$ $ \quad \Gamma, \Gamma'$ $ \quad \Gamma_1, \dots, \Gamma_n$	M M
<i>formula</i>	$::=$ $ \quad \mathbf{judgement}$ $ \quad \mathbf{formula_1 \dots formula_n}$ $ \quad \mathbf{C \neq D}$ $ \quad \mathbf{C \nlessdot D}$ $ \quad \mathbf{fields\,(C) = C_1\,f_1, \dots, C_n\,f_n}$ $ \quad \mathbf{stupid}$ $ \quad \mathbf{CT\,(C) = class_def}$ $ \quad \Gamma = \Gamma'$ $ \quad K = KK$ $ \quad \mathbf{mbody\,(m, C) = (x_1, \dots, x_n, e)}$ $ \quad 1 \leq i \leq n$	
<i>Jop</i>	$::=$ $ \quad \mathbf{x : C \in \Gamma}$	

		mtype ($\mathfrak{m}, \mathbb{C}$) = $D_1, \dots, D_n \rightarrow \mathbb{C}'$
		override ($\mathbb{C}_1, \dots, \mathbb{C}_n \rightarrow \mathbb{C}', \mathfrak{m}, D$)
		$\mathbb{C} < : D$
		$e \longrightarrow e'$
		$\Gamma \vdash e : \mathbb{C}$
		<i>class_def</i> OK
		<i>M</i> OKIN $\mathbb{C}$

judgement ::=

	<i>Jop</i>
--	------------

user_syntax ::=

	<i>ident</i>
	<i>location</i>
	<i>class_name</i>
	<i>field_name</i>
	<i>method_name</i>
	<i>constr_name</i>
	<i>index</i>
	<i>class_def</i>
	<i>constr_def</i>
	<i>method_def</i>
	<i>expr</i>
	<i>terminals</i>
	Γ
	<i>formula</i>

$\boxed{x : \mathbb{C} \in \Gamma}$

$$\frac{}{x : \mathbb{C} \in \Gamma} \text{ TIN_1}$$

$$\frac{x : \mathbb{C} \in \Gamma}{x : \mathbb{C} \in \Gamma, x' : \mathbb{C}'} \text{ TIN_2}$$

$\boxed{\text{mtype}(\mathfrak{m}, \mathbb{C}) = D_1, \dots, D_n \rightarrow \mathbb{C}'}$

$\frac{}{\text{mtype}(\mathfrak{m}, \mathbb{C}) = D_1, \dots, D_n \rightarrow \mathbb{C}} \text{ GET_PLACEHOLDERDEFINTION}$

$\boxed{\text{override}(\mathbb{C}_1, \dots, \mathbb{C}_n \rightarrow \mathbb{C}', \mathfrak{m}, D)}$

$\frac{}{\text{override}(\mathbb{C}_1, \dots, \mathbb{C}_n \rightarrow \mathbb{C}_0, \mathfrak{m}, D)} \text{ OVR_PLACEHOLDERDEFINTION}$

$\boxed{\mathbb{C} < : D}$

$$\frac{}{\mathbb{C} < : \mathbb{C}} \text{ S_REFL}$$

$$\frac{\mathbb{C} < : D \quad D < : E}{\mathbb{C} < : E} \text{ S_TRANS}$$

$\frac{\text{CT}(\mathbb{C}) = \text{class } \mathbb{C} \text{ extends } D \{ \mathbb{C}_1 \text{ f}_1, \dots, \mathbb{C}_i \text{ f}_i; K, M_1, \dots, M_j \}}{\text{<<no parses (char 2): c ***<: D >>}} \text{ S_INH}$

$$\boxed{e \longrightarrow e'}$$

$$\frac{1 \leq i \leq k \quad \mathbf{fields}(C) = C_1 f_1, \dots, C_n f_n}{\ll \text{no parses (char 2): :d***eeper: (new C(e1 , \dots, en)) .fi --> ei} \gg} \quad \text{R_FIELD}$$

$$\frac{\mathbf{mbody}(m, C) = (x_1, \dots, x_k, e)}{\ll \text{no parses (char 2): :d***eeper: (new C(e1 , \dots, en)).m(e'1, \dots, e'k) --> \{e'1/x1, \dots, e'k/xk, t\}} \gg}$$

$$\frac{C <: D}{\ll \text{no parses (char 2): :d***eeper: (D)(new C(e1, \dots, en)) --> new C(e1, \dots, en)} \gg} \quad \text{R_CAST}$$

$$\boxed{\Gamma \vdash e : C}$$

$$\frac{x : C \in \Gamma}{\Gamma \vdash x : C} \quad \text{TY_VAR}$$

$$\frac{\begin{array}{l} \Gamma \vdash e_0 : C_0 \\ \mathbf{mtype}(m, C_0) = D_1, \dots, D_n \rightarrow C \\ \Gamma \vdash e_1 : C_1 \quad \dots \quad \Gamma \vdash e_n : C_n \\ C_1 <: D_1 \quad \dots \quad C_n <: D_n \end{array}}{\Gamma \vdash e_0.m(e_1, \dots, e_n) : C} \quad \text{TY_INVK}$$

$$\frac{\begin{array}{l} \mathbf{fields}(C) = D_1 f_1, \dots, D_n f_n \\ \Gamma \vdash e_1 : C_1 \quad \dots \quad \Gamma \vdash e_n : C_n \\ C_1 <: D_1 \quad \dots \quad C_n <: D_n \end{array}}{\Gamma \vdash \mathbf{new} C(e_1, \dots, e_n) : C} \quad \text{TY_NEW}$$

$$\frac{\begin{array}{l} \Gamma \vdash e : D \\ D <: C \end{array}}{\Gamma \vdash (C)e : C} \quad \text{TY_CAST1}$$

$$\frac{\begin{array}{l} \Gamma \vdash e : D \\ C <: D \\ C \neq D \end{array}}{\Gamma \vdash (C)e : C} \quad \text{TY_CAST2}$$

$$\frac{\begin{array}{l} \Gamma \vdash e : D \\ C \not<: D \\ D \not<: C \\ \mathbf{stupid} \end{array}}{\Gamma \vdash (C)e : C} \quad \text{TY_CAST3}$$

$$\boxed{\text{class_def OK}}$$

$$\frac{\begin{array}{l} K = C(D_1 x_1, \dots, D_i x_i, C_1 y_1, \dots, C_j y_j) \{ \mathbf{super}(x_1, \dots, x_i); f_1 = y_1, \dots, f_j = y_j; \} \\ \mathbf{fields}(D) = D_1 f'_1, \dots, D_i f'_i \\ M_1 \text{ OKIN } C \quad \dots \quad M_n \text{ OKIN } C \end{array}}{\mathbf{class} C \text{ extends } D \{ C_1 f_1, \dots, C_j f_j; K, M_1, \dots, M_n \} \text{ OK}} \quad \text{MCL_1}$$

$$\boxed{M \text{ OKIN } C}$$

$$\frac{\begin{array}{l} x_1 : C_1, \dots, x_n : C_n \vdash e' : D' \\ D' <: C' \\ \mathbf{CT}(C) = \mathbf{class} C \text{ extends } D \{ C'_1 f_1, \dots, C'_i f_i; K, M_1, \dots, M_j \} \\ \mathbf{override}(C_1, \dots, C_n \rightarrow C', m, D) \end{array}}{C' m(C_1 x_1, \dots, C_n x_n) \{ \mathbf{return} e'; \} \text{ OKIN } C} \quad \text{MTY_METH}$$

Definition rules:	14 good	4 bad
Definition rule clauses:	46 good	4 bad